

Task 1:

What do you understand by exceptions?

3 min 10.20 to 10.23

Exceptions are like unexpected problems that happen when our program is running.

They're not just simple errors like typos — instead, they're things that can go wrong while the program is already running, like:

- Trying to open a file that doesn't exist

- Dividing a number by zero

- Accessing something outside the bounds of an array

- Trying to use something that's null

When one of these things happens, Java throws an exception — which is basically the program's way of saying, “Something went wrong, and I don't know how to continue unless you tell me what to do.”

If you don't handle it (using `try-catch`), your program will usually crash and stop.

But if you do handle it, you can control what happens next — like showing a helpful message or recovering in a smarter way.

Task 2:

What are the categories of Exceptions do we have in Java? What are they?

5 min 10.24 to 10.27

3 types

- Checked Exceptions

- Unchecked Exceptions (Runtime Exceptions)

- Errors

Task 3:

Can you try the below code snippet and let me know which kind of exception is this ?

What is the output of the code..?

```
// Java program to demonstrates handling
// the exception using try-catch block
import java.io.*;

class Geeks {
    public static void main(String[] args)
    {
        int n = 10;
        int m = 0;

        try {

            // Code that may throw an exception
            int ans = n / m;
            System.out.println("Answer: " + ans);
        }
        catch (ArithmeticException e) {

            // Handling the exception
            System.out.println(
                "Error: Division by zero is not allowed!");
        }
        catch (ArithmeticException e) {

            // Handling the exception
            System.out.println(
                "Error: Division by zero is not allowed!");
        }
        finally {
            System.out.println(
                "Program continues after handling the exception.");
        }
    }
}
```

5 min 10.37 to 10.42

```
Main.java  DivisionByZeroDemo.java x
1 public class DivisionByZeroDemo {
2     public static void main(String[] args) {
3
4         // Step 3: Declare two numbers
5         int n = 10;
6         int m = 0;
7
8         try {
9             // Step 4: Attempt to divide
10            int ans = n / m;
11
12            // This line won't be reached if division fails
13            System.out.println("Answer: " + ans);
14        } catch (ArithmeticException e) {
15            // Step 5: Catch the division by zero
16            System.out.println("X Error: Division by zero is not allowed!");
```

```
Project v
  DivisionByZeroDemo D:\Day8\DivisionByZeroDemo
    .idea
    out
    src
      DivisionByZeroDemo
      Main
      .gitignore
      DivisionByZeroDemo.iml
  External Libraries
  Scratches and Consoles

Main.java  DivisionByZeroDemo.java x
1 public class DivisionByZeroDemo {
2     public static void main(String[] args) {
3
4         int ans = n / m;
5
6         // This line won't be reached if division fails
7         System.out.println("Answer: " + ans);
8     } catch (ArithmeticException e) {
9         // Step 5: Catch the division by zero
10        System.out.println("X Error: Division by zero is not allowed!");
11    } finally {
12        // Step 6: This block always runs
13        System.out.println("Program continues after handling the exception.");
14    }
15 }
16
17
18
19
20
21
22
23

Run DivisionByZeroDemo x
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.jar=50977" -Dfile.en
X Error: Division by zero is not allowed!
Program continues after handling the exception.
Process finished with exit code 0
```

Task 4:

List of checked and unchecked exceptions.

Checked Exceptions

(Handled at compile-time – you must catch or declare them with throws)

Exception	Description
-----------	-------------

IOException	Input/output failures
FileNotFoundException	File not found during input/output
SQLException	Database access errors
ParseException	Parsing issues (e.g. date parsing)
ClassNotFoundException	Requested class not found
InterruptedException	Thread interruption
NoSuchMethodException	Method does not exist via reflection
InstantiationException	Object cannot be instantiated via reflection
InvocationTargetException	Exception thrown by an invoked method

Unchecked Exceptions

(Handled at runtime – optional to catch)

Exception	Description
ArithmeticException	Illegal arithmetic (e.g. divide by zero)
NullPointerException	Accessing object with a null reference
ArrayIndexOutOfBoundsException	Accessing invalid index in an array

StringIndexOutOfBoundsException	Invalid index in a string
IllegalArgumentException	Passing an illegal argument to a method
NumberFormatException	Invalid conversion from string to number
ClassCastException	Invalid type casting
IllegalStateException	Inappropriate method call for current state
IndexOutOfBoundsException	General index error (lists, arrays, etc.)

Task 5:

Try with Multiple catch blocks Execute the below code snippet n display the out .. along with reason..

```
public class ExcepTest {

    public static void main(String args[]) {
        try {
            int a[] = new int[2];
            int b = 0;
            int c = 1/b;
            System.out.println("Access element three : " + a[3]);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("ArrayIndexOutOfBoundsException thrown : " + e);
        }
        catch (Exception e) {
            System.out.println("Exception thrown : " + e);
        }
    }
}
```

```

        System.out.println("Out of the block");
    }
}

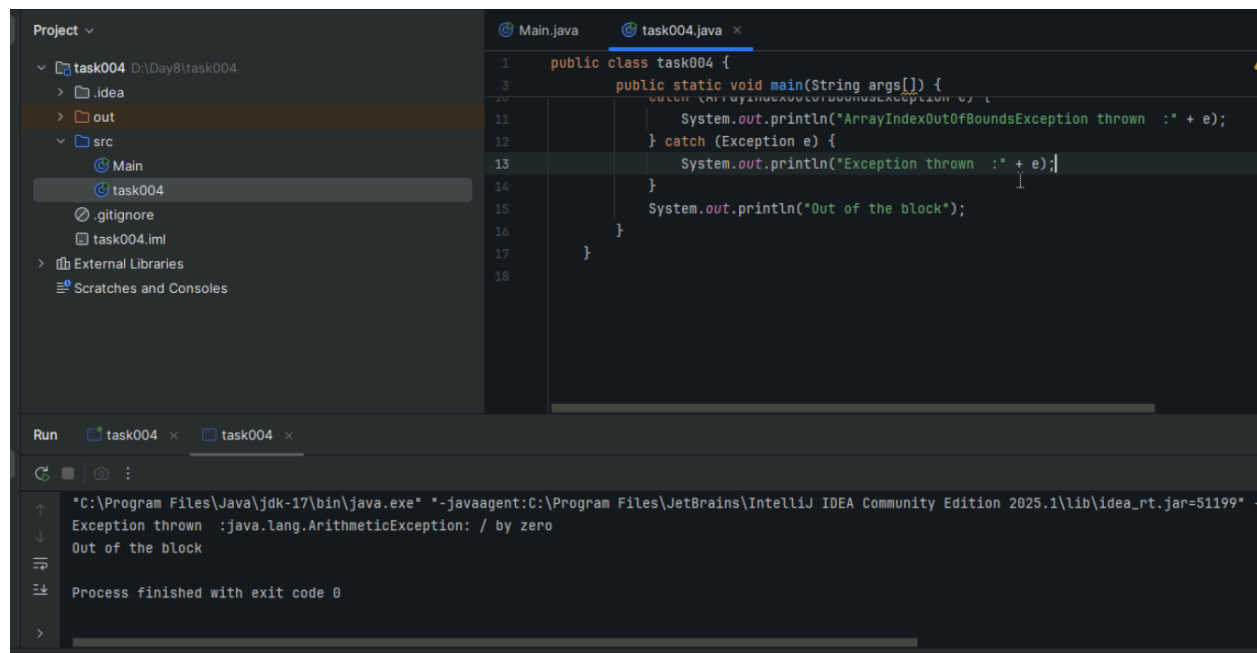
```

```

public class task004 {

    public static void main(String args[]) {
        try {
            int a[] = new int[2]; // array of size 2: indices 0,1
            int b = 0;
            int c = 1/b;           // causes ArithmeticException (divide by zero)
            System.out.println("Access element three : " + a[3]); // would cause ArrayIndexOutOfBoundsException
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("ArrayIndexOutOfBoundsException thrown : " + e);
        } catch (Exception e) {
            System.out.println("Exception thrown : " + e);
        }
        System.out.println("Out of the block");
    }
}

```



Task 6:

What is the output of the below code... give your reason for the output

```

public class ExcepTest {

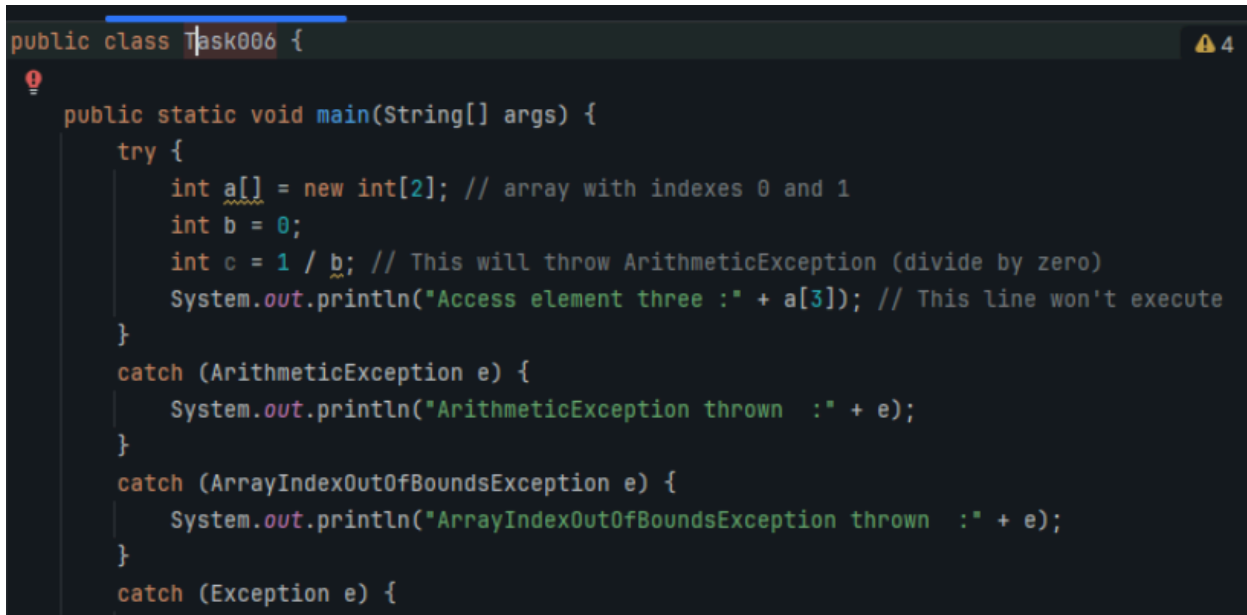
    public static void main(String args[]) {

```

```

try {
    int a[] = new int[2];
    int b = 0;
    int c = 1/b;
    System.out.println("Access element three :" + a[3]);
}
catch (ArithmeticException e) {
    System.out.println("ArithmeticException thrown :" + e);
}
catch (ArrayIndexOutOfBoundsException e) {
    System.out.println("ArrayIndexOutOfBoundsException thrown :" + e);
}
catch (Exception e) {
    System.out.println("Exception thrown :" + e);
}
System.out.println("Out of the block");
}
}

```



```

public class Task006 {
    public static void main(String[] args) {
        try {
            int a[] = new int[2]; // array with indexes 0 and 1
            int b = 0;
            int c = 1 / b; // This will throw ArithmeticException (divide by zero)
            System.out.println("Access element three :" + a[3]); // This line won't execute
        }
        catch (ArithmeticException e) {
            System.out.println("ArithmeticException thrown :" + e);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("ArrayIndexOutOfBoundsException thrown :" + e);
        }
        catch (Exception e) {

```

The screenshot shows the IntelliJ IDEA IDE with a project named 'task006'. The source code is in a file named 'Task006.java'. The code defines a public class 'Task006' with a static method 'main' that takes a String array 'args'. Inside the 'main' method, there are three catch blocks: 'ArithmeticException', 'ArrayIndexOutOfBoundsException', and 'Exception'. Each catch block prints a message to the console. The 'main' method also prints 'Out of the block' after the catch blocks. The Run tab shows the execution output: 'ArithmeticException thrown :java.lang.ArithmeticException: / by zero' and 'Out of the block'. The process finished with exit code 0.

```
public class Task006 {
    public static void main(String[] args) {
        catch (ArithmeticException e) {
            System.out.println("ArithmeticException thrown : " + e);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("ArrayIndexOutOfBoundsException thrown : " + e);
        }
        catch (Exception e) {
            System.out.println("Exception thrown : " + e);
        }
        System.out.println("Out of the block");
    }
}
```

Run Task006 x

"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.jar=51377" -

ArithmeticException thrown :java.lang.ArithmeticException: / by zero

Out of the block

Process finished with exit code 0

Task 7:

In the below code we are having use multiple catch in a single statement: find the output and try to understand the code..

```
public class ExcepTest {

    public static void main(String args[]) {
        try {
            int a[] = new int[2];
            int b = 0;
            int c = 1/b;
            System.out.println("Access element three : " + a[3]);
        }
        catch (ArrayIndexOutOfBoundsException | ArithmeticException e) {
            System.out.println("Exception thrown : " + e);
        }
        System.out.println("Out of the block");
    }
}
```

11.15 to 11.18

Naested try blocks


```

public class Task007 {

    public static void main(String[] args) {
        try {
            int a[] = new int[2]; // array with indexes 0 and 1
            int b = 0;
            int c = 1 / b;        // This line throws ArithmeticException
            System.out.println("Access element three : " + a[3]); // Won't be executed
        }

        // Multi-catch block: handles both exceptions
        catch (ArrayIndexOutOfBoundsException | ArithmeticException e) {
            System.out.println("Exception thrown : " + e);
        }

        System.out.println("Out of the block");
    }
}

```

The screenshot shows the IntelliJ IDEA IDE. On the left is the Project Explorer with a tree view containing folders like 'out' and 'src', and files like 'Main', 'Task007', '.gitignore', and 'Task007.iml'. The main editor window displays the Java code for 'Task007'. The code is identical to the one in the first block. Below the editor is the Run tool window, which shows the output of the program. The output consists of three lines: 'Exception thrown :java.lang.ArithmeticException: / by zero', 'Out of the block', and 'Process finished with exit code 0'.

```

3      public static void main(String[] args) {
9          }
10
11         // Multi-catch block: handles both exceptions
12         catch (ArrayIndexOutOfBoundsException | ArithmeticException
13             System.out.println("Exception thrown : " + e);
14         }
15
16         System.out.println("Out of the block");
17     }
18 }
19

```

Task007 x

"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\
 Exception thrown :java.lang.ArithmeticException: / by zero
 Out of the block
 Process finished with exit code 0

Task 008:

```

public class ExcepTest {

    public static void main(String args[]) {

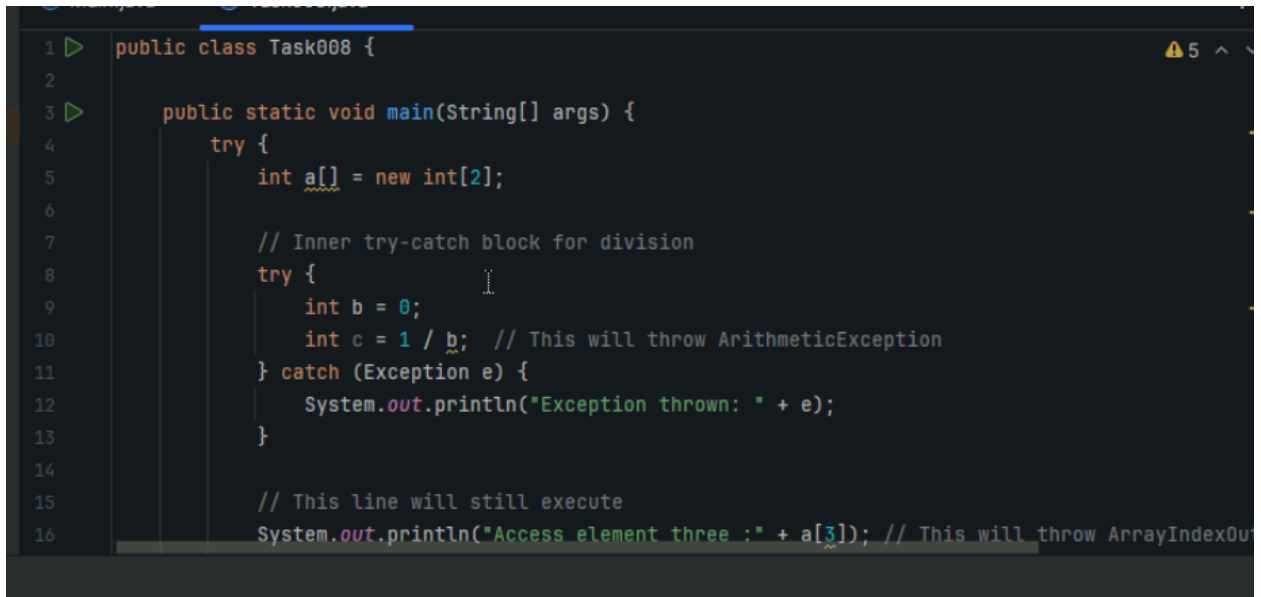
```

```

try {
    int a[] = new int[2];
    try {
        int b = 0;
        int c = 1/b;
    } catch (Exception e) {
        System.out.println("Exception thrown: " + e);
    }
    System.out.println("Access element three : " + a[3]);
}
catch (ArrayIndexOutOfBoundsException e) {
    System.out.println("Exception thrown: " + e);
}
System.out.println("Out of the block");
}
}

```

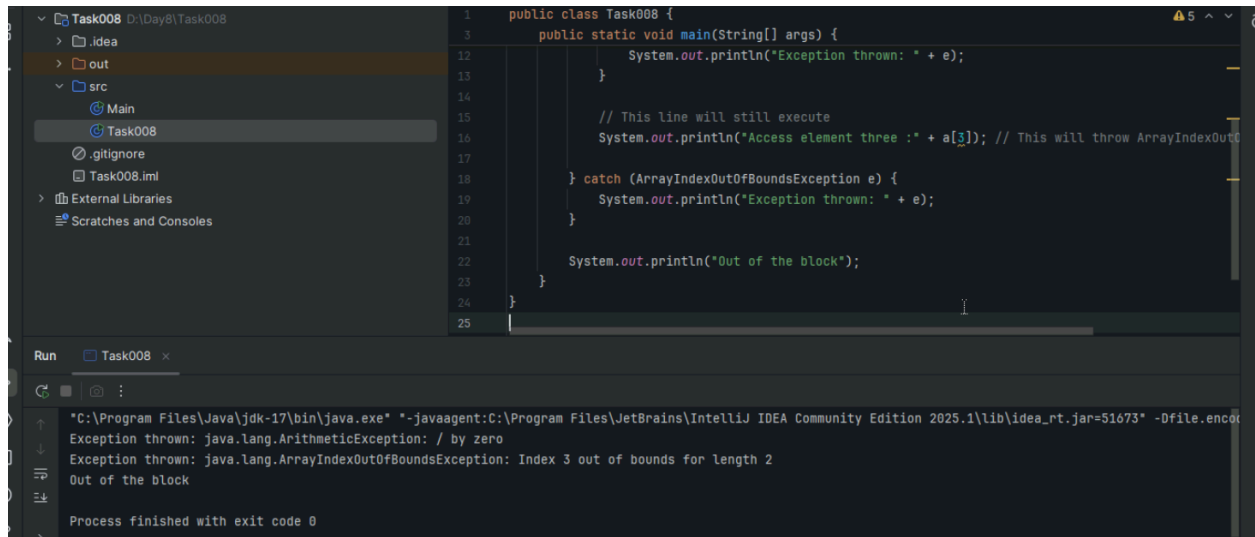
Throw and Throws:



```

1  public class Task008 {
2
3  public static void main(String[] args) {
4      try {
5          int a[] = new int[2];
6
7          // Inner try-catch block for division
8          try {
9              int b = 0;
10             int c = 1 / b; // This will throw ArithmeticException
11         } catch (Exception e) {
12             System.out.println("Exception thrown: " + e);
13         }
14
15         // This line will still execute
16         System.out.println("Access element three : " + a[3]); // This will throw ArrayIndexOut

```



Task 009

// Demonstrating how to throw an exception

```
class MyClass {
```

```
    static void fun() throws IllegalAccessException
    {
        System.out.println("Inside fun(). ");
        throw new IllegalAccessException("demo");
    }
}
```

```
public static void main(String args[])
```

```
{
    try {
        fun();
        method2(); → arrayindex...
        Method3() → file not found....
    }
    catch (IllegalAccessException e) {
        System.out.println("Caught in main.");
    }
}
}
```

```
class Task009 {

    static void fun() throws IllegalAccessException 1 usage
    {
        System.out.println("Inside fun(). ");
        throw new IllegalAccessException("demo");
    }

    public static void main(String args[])
    {
        try {
            fun();
            // method2(); → arrayindex...
            // Method3() → file not found...
        }
        catch (IllegalAccessException e) {
```

```
12         fun();
13         // method2(); → arrayindex...
14         // Method3() → file not found...
15     }
16     catch (IllegalAccessException e) {
17         System.out.println("Caught in main.");
18     }
19 }
20
21
22
23
24
25
```

Task009 x

```
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.jar=51875"
Inside fun().
Caught in main.

Process finished with exit code 0
```

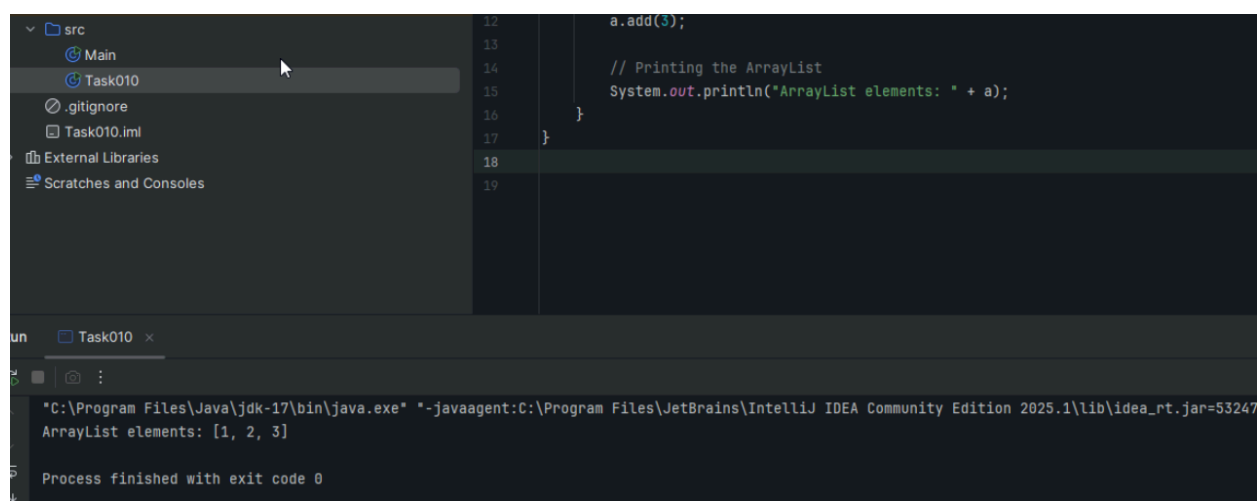
Task10

```
import java.util.ArrayList;
```

```
class Main {
```

```
public static void main (String[] args) {  
  
    // Creating an ArrayList  
    ArrayList<Integer> a = new ArrayList<Integer>();  
  
    // Adding Element in ArrayList  
    a.add(1);  
    a.add(2);  
    a.add(3);  
  
    // Printing ArrayList  
    System.out.println(a);  
}  
}
```

```
1 import java.util.ArrayList;
2
3 public class Task010 {
4     public static void main(String[] args) {
5
6         // Creating an ArrayList of Integer type
7         ArrayList<Integer> a = new ArrayList<Integer>();
8
9         // Adding elements to the ArrayList
10        a.add(1);
11        a.add(2);
12        a.add(3);
13
14        // Printing the ArrayList
15        System.out.println("ArrayList elements: " + a);
16    }
17 }
18
19
```



The screenshot shows the IntelliJ IDEA IDE. The left sidebar displays the project structure with 'src' containing 'Main' and 'Task010'. The 'Task010' file is selected. The main editor shows the Java code for 'Task010'. The code creates an 'ArrayList' of 'Integer' type, adds three elements (1, 2, 3), and prints the list. The bottom console shows the execution output: 'ArrayList elements: [1, 2, 3]' and 'Process finished with exit code 0'.

Task 011

Wap to create an array list to display 10 elements using for loop.

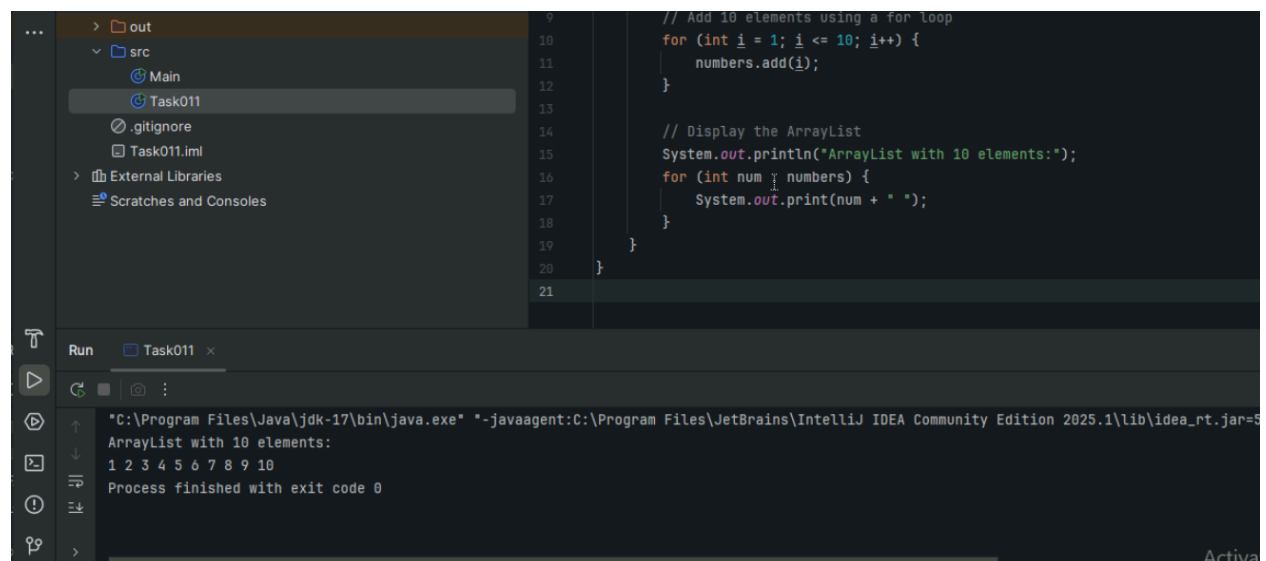
```
import java.util.ArrayList;

> public class Task011 {
>     public static void main(String[] args) {

        // Create an ArrayList of Integer type
        ArrayList<Integer> numbers = new ArrayList<>();

        // Add 10 elements using a for loop
        for (int i = 1; i <= 10; i++) {
            numbers.add(i);
        }

        // Display the ArrayList
        System.out.println("ArrayList with 10 elements:");
        for (int num : numbers) {
```



Task 012

Find the output of the be code snippet..

// Addition, Deletion and Updation of Element

```
import java.util.*;
```

```
class Main {
```

```
    public static void main(String args[]){
```

```
        ArrayList<String> al = new ArrayList<>();
```

```
        al.add("Prasunamba");
```

```
        al.add("Meher");
```

```
        System.out.println("Original List : "+al);
```

```
        al.add(1, "Hello");
```

```
        System.out.println("After Adding element at index 1 : "+ al);
```

```
        al.remove(0);
```

```
        System.out.println("Element removed from index 0 : "+ al);
```

```
        al.remove("Prasunamba");
```

```
        System.out.println("Element Prasunamba removed : "+ al);
```

```
        al.set(0, "K");
```



```
System.out.println("List after updation of value : "+al);
```

```
}
```

```
}
```

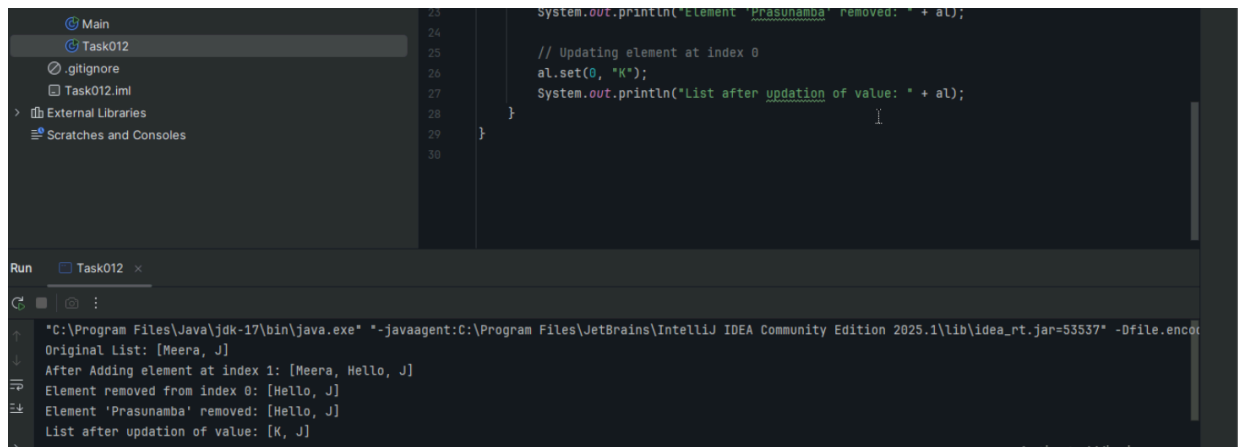
```
Main.java Task012.java
import java.util.ArrayList;

public class Task012 {
    public static void main(String[] args) {
        // Creating an ArrayList of Strings
        ArrayList<String> al = new ArrayList<>();

        // Adding initial elements
        al.add("Meera");
        al.add("J");
        System.out.println("Original List: " + al);

        // Adding an element at index 1
        al.add(index: 1, element: "Hello");
        System.out.println("After Adding element at index 1: " + al);
```

```
3 public class Task012 {
4     public static void main(String[] args) {
15         System.out.println("After Adding element at index 1: " + al);
16
17         // Removing element at index 0
18         al.remove(index: 0);
19         System.out.println("Element removed from index 0: " + al);
20
21         // Removing element by value
22         al.remove(o: "Prasunamba");
23         System.out.println("Element 'Prasunamba' removed: " + al);
24
25         // Updating element at index 0
26         al.set(0, "K");
27         System.out.println("List after updation of value: " + al);
28     }
```

A screenshot of an IDE window. The top pane shows Java code with line numbers 24 to 30. The code includes a print statement for 'Prasunamba', a comment about updating an element, a set operation, and another print statement. The bottom pane shows the output of the program, which includes the original list, the list after adding an element, the list after removing an element, and the list after updating a value.

```
24 System.out.println("Element 'Prasunamba' removed: " + al);
25
26 // Updating element at index 0
27 al.set(0, "K");
28 System.out.println("List after updation of value: " + al);
29 }
30
```

Run Task012 x

```
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.jar=53537" -Dfile.encoding=UTF-8
Original List: [Meera, J]
After Adding element at index 1: [Meera, Hello, J]
Element removed from index 0: [Hello, J]
Element 'Prasunamba' removed: [Hello, J]
List after updation of value: [K, J]
```

Task 013

Run the code and see how the user defined exception works..

User defined Exception:

// A Class that represents user-defined exception

class Customer extends Exception { // predefined class Exception

public Customer(String m) { // constructor with parameters

super(m); // parent class constructor

}

}

// A Class that uses the above MyException

public class setText {

public static void main(String args[]) {

try {

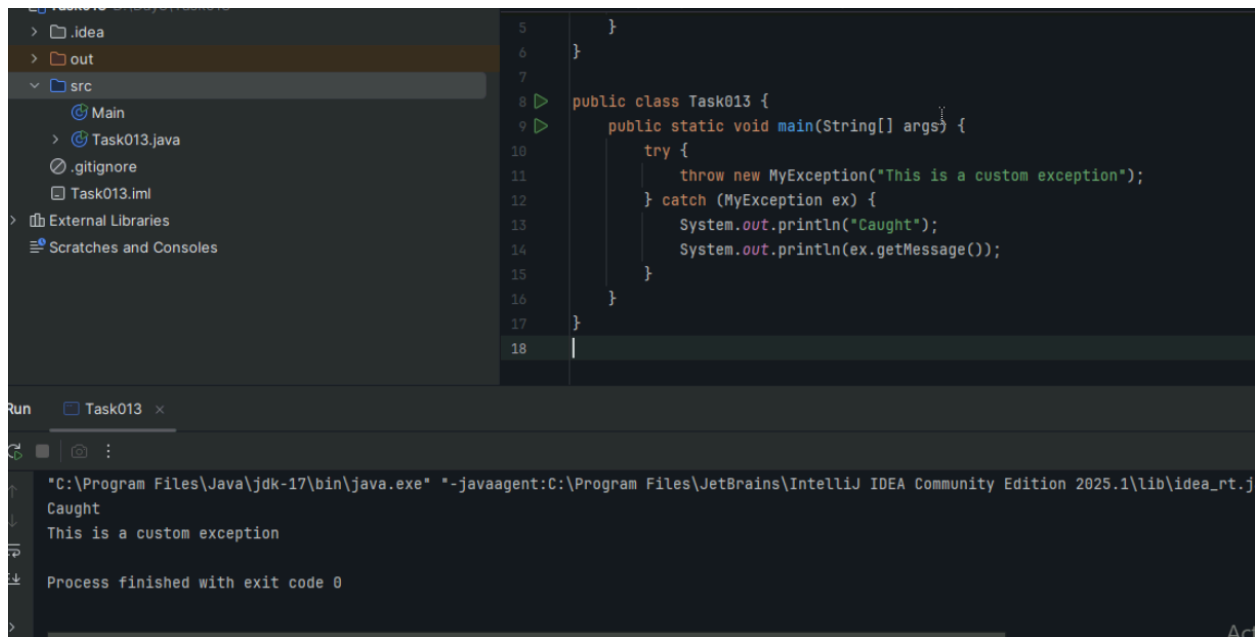
// Throw an object of user-defined exception

```
        throw new MyException("This is a custom exception");
    }
    catch (MyException ex) {
        System.out.println("Caught");
        System.out.println(ex.getMessage());
    }
}
}
```

15.26 to 15.30 4 min

```
// User-defined exception class
class MyException extends Exception { 2 usages
    public MyException(String message) { 1 usage
        super(message);
    }
}

public class Task013 {
    public static void main(String[] args) {
        try {
            throw new MyException("This is a custom exception");
        } catch (MyException ex) {
            System.out.println("Caught");
            System.out.println(ex.getMessage());
        }
    }
}
```



```
5    }
6  }
7
8  public class Task013 {
9      public static void main(String[] args) {
10         try {
11             throw new MyException("This is a custom exception");
12         } catch (MyException ex) {
13             System.out.println("Caught");
14             System.out.println(ex.getMessage());
15         }
16     }
17 }
18
```

Run Task013

```
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.j
Caught
This is a custom exception

Process finished with exit code 0
```

Task 014

Inheritance

Classes customer/ person , employee, Manager ... 2 variables in each class

Driver class – display all the variables... toString();

Hint : use getter and setters..

8 min 17.20 to 17.28

Inner classes

```
// Employee class extends Customer
class Employee extends Customer { no usages 1 inheritor
    private String department; 4 usages
    private double salary; 4 usages

    public Employee(String name, int id, String department, double salary) { no usages
        super(name, id);
        this.department = department;
        this.salary = salary;
    }

    public String getDepartment() { no usages
        return department;
    }

    public void setDepartment(String department) { no usages
        this.department = department;
    }
}
```

```
    public void setDepartment(String department) { no usages
        this.department = department;
    }

    public double getSalary() { no usages
        return salary;
    }

    public void setSalary(double salary) { no usages
        this.salary = salary;
    }

    public String toString() { 1 override
        return super.toString() + ", Department: " + department + ", Salary: " + salary;
    }
}
```

The screenshot shows the IntelliJ IDEA interface. On the left, the Project Structure panel displays a hierarchy: Employee, Main, Manager, Task014, Task0014.java, Customer, .gitignore, Task0014.iml, External Libraries, and Scratches and Consoles. The main editor window shows the Employee class code, specifically the toString() method. Below the editor, the Run tab is active, showing the command: "C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.jar=57268" -Dfile.encoding=UTF-8. The output of the run is displayed in the console: Customer Details: Customer Name: Anjali, ID: 101. Employee Details: Customer Name: Ravi, ID: 102, Department: IT, Salary: 55000.0.

```
Customer Details:
Customer Name: Anjali, ID: 101

Employee Details:
Customer Name: Ravi, ID: 102, Department: IT, Salary: 55000.0
```

```
// Manager class extends Employee
class Manager extends Employee { no usages
    private int teamSize; 4 usages
    private String level; 4 usages

    public Manager(String name, int id, String department, double salary, int teamSize, String
        super(name, id, department, salary);
        this.teamSize = teamSize;
        this.level = level;
    }

    public int getTeamSize() {
        return teamSize;
    }

    public void setTeamSize(int teamSize) { no usages
        this.teamSize = teamSize;
    }
}
```

Manager
private int teamSize
Task0014

```
    public void setTeamSize(int teamSize) { no usages
        this.teamSize = teamSize;
    }

    public String getLevel() { no usages
        return level;
    }

    public void setLevel(String level) { no usages
        this.level = level;
    }

    public String toString() {
        return super.toString() + ", Team Size: " + teamSize + ", Level: " + level;
    }
}
```

```

    }

    public void setLevel(String level) { no usages
        this.level = level;
    }

    public String toString(){
        return super.toString() + ", Team Size: " + teamSize + ", Level: " + level;
    }
}

```

```

class Customer { 1 usage 2 inheritors
    private String name; 4 usages
    private int id; 4 usages

    public Customer(String name, int id) { 1 usage
        this.name = name;
        this.id = id;
    }

    // Getters and setters
    public String getName() { no usages
        return name;
    }

    public void setName(String name) { no usages
        this.name = name;
    }
}

```

```

    public void setName(String name) { no usages
        this.name = name;
    }

    public int getId() { no usages
        return id;
    }

    public void setId(int id) { no usages
        this.id = id;
    }

    public String toString() { 2 overrides
        return "Customer Name: " + name + ", ID: " + id;
    }
}

```

```

// Driver class
public class Task014 {
    public static void main(String[] args) {

        // Creating objects
        Customer customer = new Customer(name: "Anjali", id: 101);
        Employee employee = new Employee(name: "Ravi", id: 102, department: "IT", salary: 55000);
        Manager manager = new Manager(name: "Neha", id: 103, department: "HR", salary: 75000, teamS

        // Displaying all objects using toString()
        System.out.println("Customer Details:");
        System.out.println(customer);

        System.out.println("\nEmployee Details:");
        System.out.println(employee);
    }
}

```



```

public class Task014 {
    public static void main(String[] args) {
        System.out.println(customer);

        System.out.println("\nEmployee Details:");
        System.out.println(employee);

        System.out.println("\nManager Details:");
        System.out.println(manager);
    }
}

```

```

"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files
Customer Details:
Customer Name: Anjali, ID: 101

Employee Details:
Customer Name: Ravi, ID: 102, Department: IT, Salary: 55000.0

```

```

Manager Details:
Customer Name: Neha, ID: 103, Department: HR, Salary: 75000.0, Team Size: 5, Level: Senior

Process finished with exit code 0

```

Inner classes

Task 015

What is the output of the below code snippet.. Explain ..

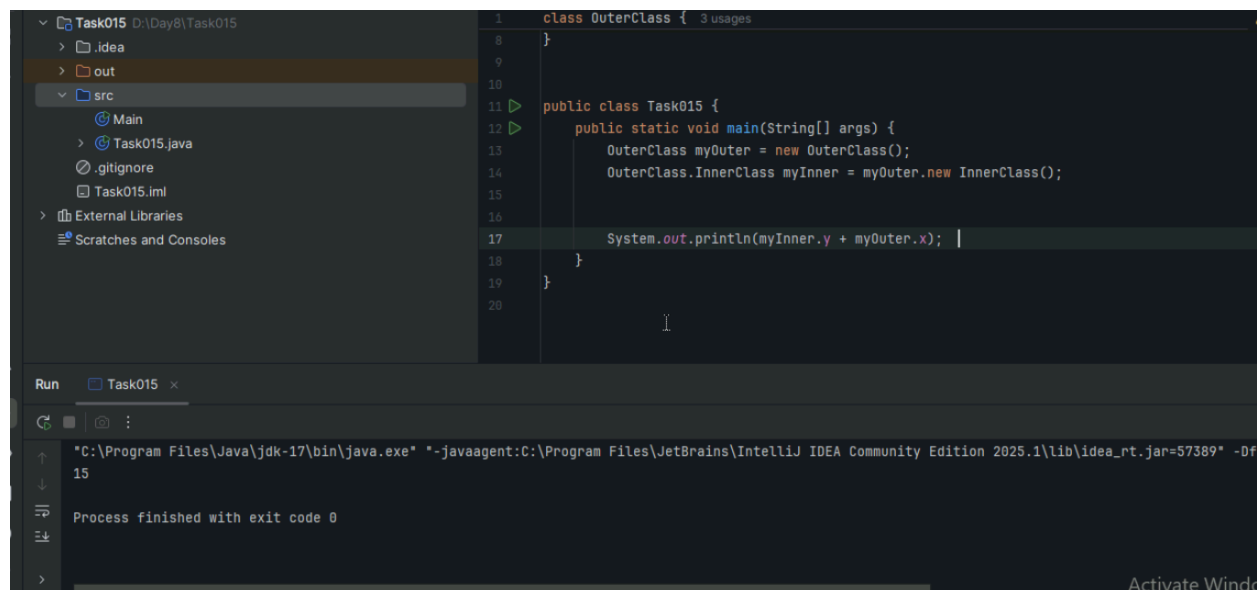
```
class OuterClass {  
  
    int x = 10;  
  
    class InnerClass {  
  
        int y = 5;  
  
    }  
}
```

```
public class Main {  
  
    public static void main(String[] args) {  
  
        OuterClass myOuter = new OuterClass();  
  
        OuterClass.InnerClass myInner = myOuter.new InnerClass();  
  
        System.out.println(myInner.y + myOuter.x);  
  
    }  
}
```

```
class OuterClass { 3 usages
    int x = 10; 1 usage

    class InnerClass { 2 usages
        int y = 5; 1 usage
    }
}

public class Task015 {
    public static void main(String[] args) {
        OuterClass myOuter = new OuterClass();
        OuterClass.InnerClass myInner = myOuter.new InnerClass();
    }
}
```



Step 1: Define Outer and Inner Class

```
class OuterClass {
```

```
    int x = 10;
```

```
class InnerClass {  
    int y = 5;  
}  
}
```

OuterClass has a variable x with value 10.

Inside it, there's a non-static inner class called InnerClass, which has a variable y = 5.

Step 2: Create Outer Class Object

```
OuterClass myOuter = new OuterClass();
```

We create an object myOuter of OuterClass.

Step 3: Create Inner Class Object

```
OuterClass.InnerClass myInner = myOuter.new InnerClass();
```

Since InnerClass is non-static, it must be tied to an object of the outer class.

So we use myOuter.new InnerClass() to create the InnerClass object.

Step 4: Access Variables and Add

```
System.out.println(myInner.y + myOuter.x);
```

myInner.y is 5

myOuter.x is 10

Adding them: $5 + 10 = 15$

Task 016

Use the above code and make the inner class as private and see the output..

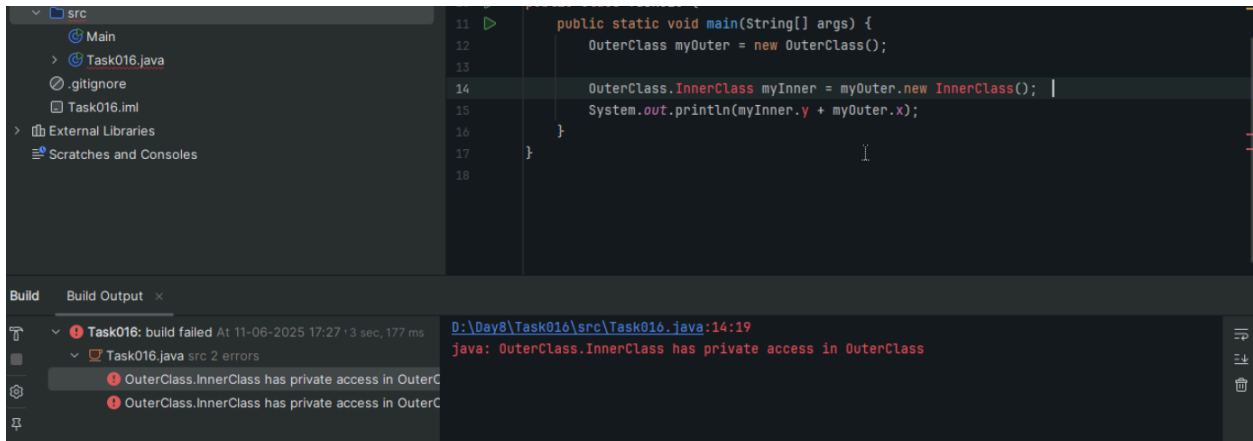
Ex: Private class InnerClass {

```
class OuterClass { 3 usages
    int x = 10; 1 usage

    private class InnerClass { no usages
        int y = 5; no usages
    }
}

> public class Task016 {
>     public static void main(String[] args) {
        OuterClass myOuter = new OuterClass();

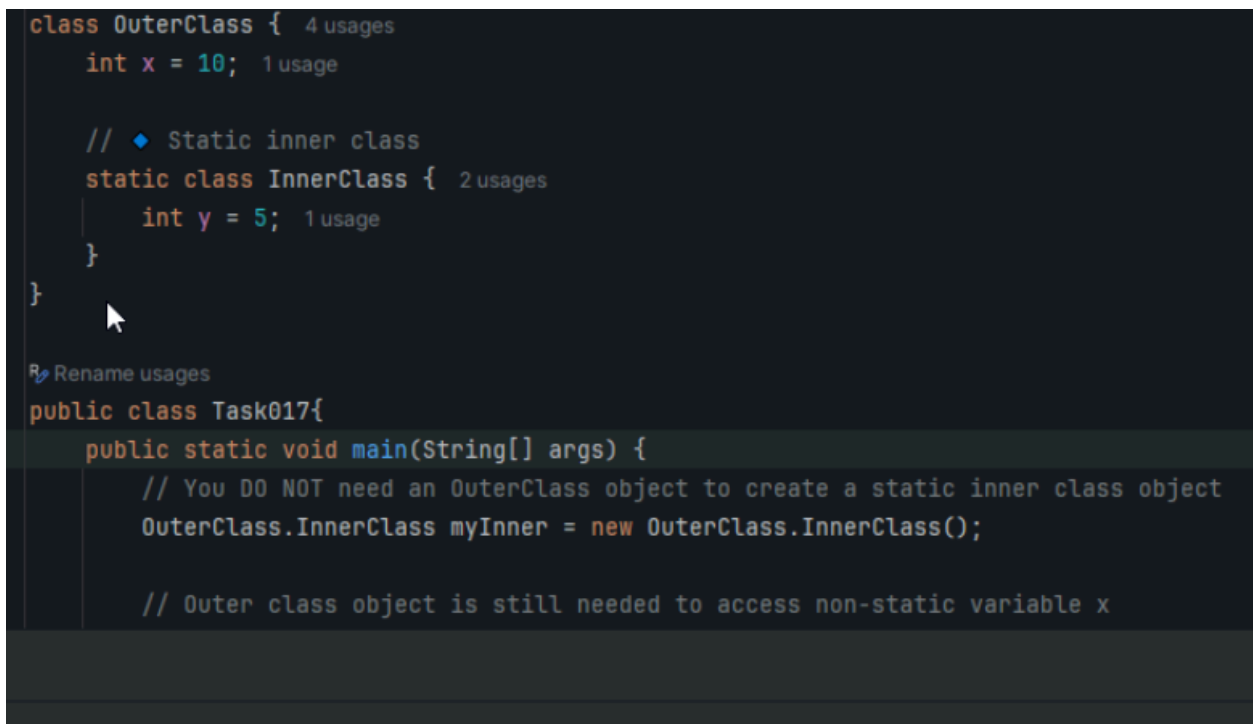
        OuterClass.InnerClass myInner = myOuter.new InnerClass(); |
        System.out.println(myInner.y + myOuter.x);
    }
}
```

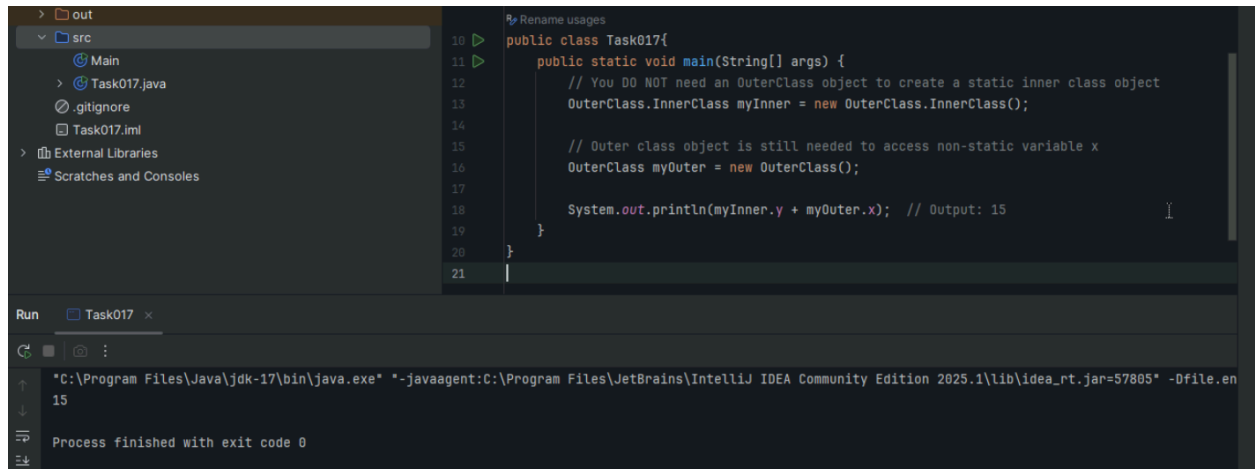


Task 017

Use the above code Task 015 and make the inner class static ... see the output and explain..

Ex: static class InnerClass {





Task 018

Use the above code Task 015 and create a method in innerclass and return the outer class variable

```
class OuterClass{  
  
    Int x = 50;  
  
    Class InnerClass {  
  
        Public int innerMethod() {  
  
            Return x;  
  
        }  
  
    }  
  
}
```

The screenshot displays the IntelliJ IDEA IDE interface. On the left, the Project Explorer shows the file structure for 'Task018', including folders '.idea', 'out', and 'src', and files 'Main', 'Task018.java', '.gitignore', and 'Task018.iml'. The main editor window shows the following Java code:

```
1 class OuterClass { 3 usages
2     int x = 50; 1 usage
3     class InnerClass { 2 usages
4         public int innerMethod() { 1 usage
5             return x;
6         }
7     }
8
9     public class Task018 {
10         public static void main(String[] args) {
11             OuterClass outer = new OuterClass();
12             OuterClass.InnerClass inner = outer.new InnerClass();
13             int result = inner.innerMethod();
14             System.out.println("Returned from innerMethod: " + result); |
15         }
16     }
```

Below the code editor, the Run tab is active, showing the command executed and the output:

```
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1\lib\idea_rt.jar=58073"
Returned from innerMethod: 50
Process finished with exit code 0
```